INCH-POUND

MIL-DTL-55302/125D
18 February 2005
SUPERSEDING
MIL-C-55302/125C(USAF)
6 MAY 1993

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, 80 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.050 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

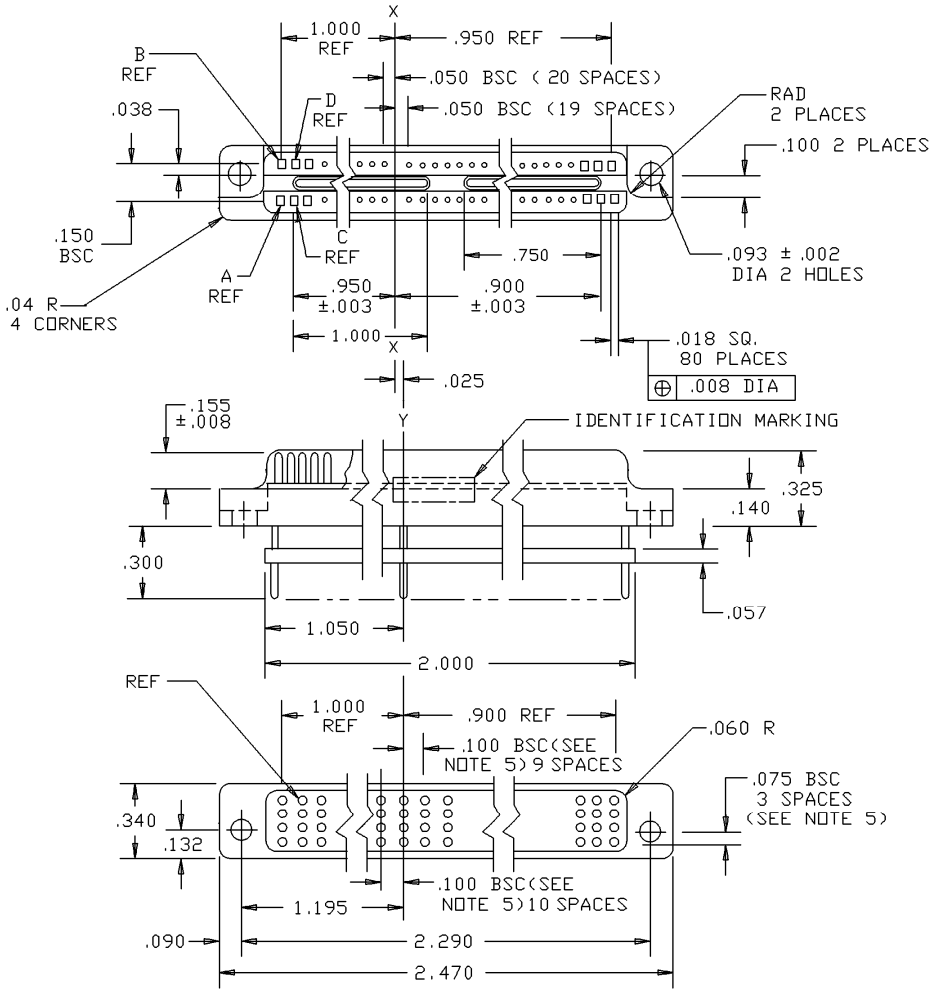
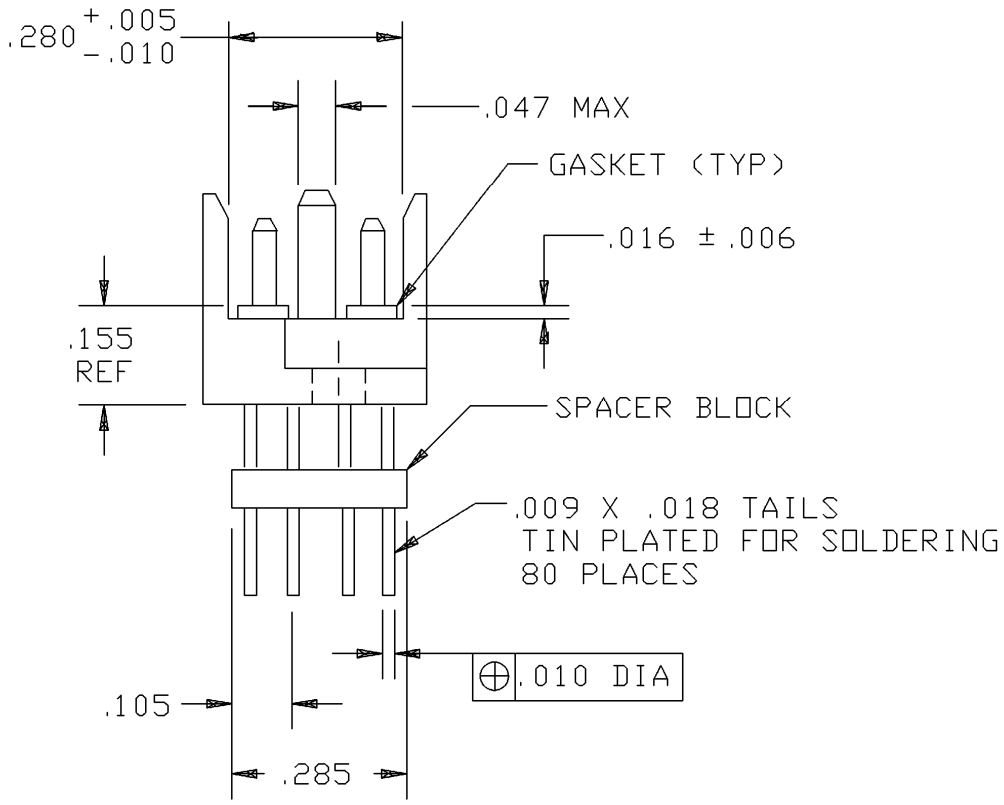


FIGURE 1. Connectors, plug, .050 (1.27 mm) spacing.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.038	0.97	.132	3.35	.900	22.86
.003	0.08	.040	1.02	.140	3.56	.950	24.13
.005	0.13	.047	1.19	.150	3.81	1.000	25.40
.006	0.15	.050	1.27	.155	3.94	1.050	26.67
.008	0.20	.057	1.45	.280	7.11	1.195	30.35
.009	0.23	.060	1.52	.285	7.24	2.000	50.80
.010	0.25	.090	2.29	.300	7.62	2.290	58.17
.016	0.41	.093	2.36	.325	8.26	2.470	62.74
.018	0.46	.100	2.54	.340	8.64		
.025	0.64	.105	2.67	.750	19.05		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm) and ± 2° on angles.
4. These connectors mate with connectors specified in MIL-DTL-55302/126.
5. Dimension applies at point of egress from spacer block.

FIGURE 1. Connectors, plug, .050 (1.27 mm) spacing – Continued.

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REQUIREMENTS

Design and construction:

Dimensions and configurations: See figure 1.

Material:

Body: In accordance with MIL-DTL-55302.

Plating: Contact plating shall be gold in accordance with MIL-DTL-55302 over nickel in accordance with MIL-DTL-55302 in the contact engagement area for .100 inch minimum length. The termination area plating shall be tin lead in accordance with MIL-DTL-55302 over nickel in accordance with MIL-DTL-55302 for .250 inch minimum length.

Contact identification: See figure 1.

Contact rating: 1.5 amperes per contact, maximum.

Operating temperature: -65°C to +125°C.

Keying: See MIL-DTL-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .30 and the withdrawal force in pounds shall be a minimum of .03 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .025 ohm, and no individual contact pair shall have resistance exceeding .030 ohm.

Contact retention: .75 pound, minimum.

Dielectric withstanding voltage:

Sea level: 450 volts rms.

High altitude: 150 volts rms.

Solderability: Each connector shall be subjected to method 208 of MIL-STD-202 (solder termination only) (applicable to qualification and retention of qualification only).

Part or Identifying Number (PIN): M55302/125-01.

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/31
MIL-DTL-55302/126
MIL-STD-202

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Previous changes. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Air Force – 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Air Force – 19, 99

(Project 5935-4477-000)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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